

ALLIANCE GIRLS 2025 PREDICTION



POSSIBLE KCSE EXAM

451/1

COMPUTER STUDIES

PAPER 1

TIME: 2½ HOURS



NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTIONS TO CANDIDATES

- Write your **Name**, **Class**, **Admission Number** and **Index Number** in the spaces provided.
- Answer **ALL** the questions in **section A**
- Answer **question 16 (Compulsory)** and **any other three** questions from **section B**.

Question	Maximum score	Actual score
1-16		
17		
18		
19		
20		
TOTAL SCORE		

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SECTION A: (40 MARKS)

Answer ALL Questions in This Section

1. During a visit to the typing room, Jane a computer student observed a type of printer that uses a beam to create an image on a rotating drum. As the beam hits the drum, some regions attract ink toner particles. The toner is then fused onto a piece of paper.

I. State the name of this type of printer and indicate whether it is an impact or non-impact printer.

(1mark)

II. List **two** advantages of using the printer mentioned in a. above.

(2marks)

2. Bunam Water Company sends out field officers to take consumer meter readings. The data collected by the officers is then keyed into the computer. The system then generates utility bills which are printed and sent to the consumers.

i. State **two** transcription errors that are likely to occur during meter reading or input.

(1 mark)

ii. State **two** ways such errors can be avoided.

(1 mark)

3. Highlight the dangers associated with the following in a computer laboratory set up:

a. Purchasing software before assessing requirements. **(1 mark)**

b. Using foreign flash disks on the computer.

(1 mark)

4. To process an instruction, the CPU goes through a cycle of three stages. Name each of the stages. **(3 marks)**

5. Explain the importance of the following spreadsheet operations. **(2 marks)**

(i) Freezing columns

(ii) Data validation

(b) Name **two** types of relationships that can be applied in database design.

(1 mark)

6. Briefly explain any Four block operations that can be performed on data in word processing

(4 marks)

7. (a) Mutiso wanted to upgrade the RAM of his computer. State two factors Mutiso should consider when purchasing a RAM module.

(1 mark)

(b) Define the following terms:

(2 marks)

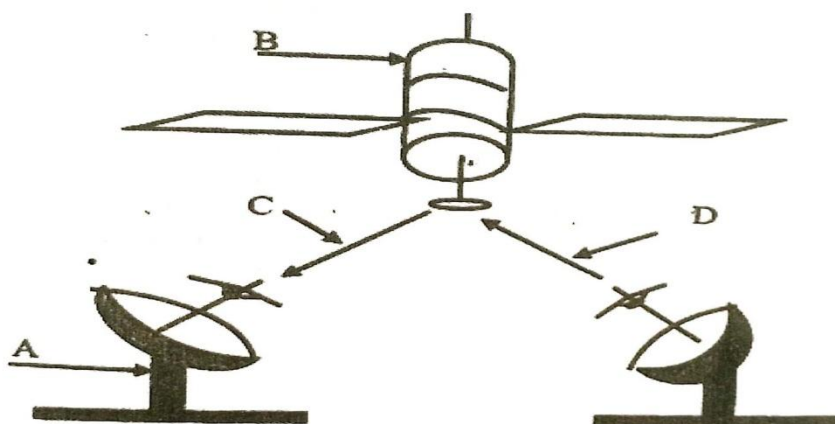
(i) Computer Port

(ii) Power supply unit

8. Mkulima Bora is a society that embraced E-Commerce recently. The manager is convinced that there is illegal access to the company's system. State TWO ways in which the society can overcome this problem.

(2 marks)

9. (a) Study the diagram below and answer the questions that follow.



(i) Name the communication media depicted in the above diagram (1 mark)

(ii) Name the parts labelled A, B, C, and D (3 marks)

A _____
B _____
C _____
D _____

10. In a desktop publishing document, an image embedded can be transformed in various ways.

Name any **three** ways one can use to make the image fit in a designated area. (3 marks)

11. Differentiate between source program and object program. (2 marks)

12. Differentiate between the following terms in Internet (4 marks)

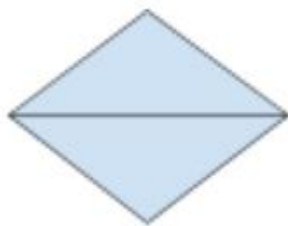
(i) Drafts and outbox

(ii) cc and bcc

13. State two characteristics of a computer that is infected by computer viruses. (2 marks)

14. Identify each of the following symbols as used in system flowchart. (1 mark)

(a)



(b)



15. Differentiate between bullets and numbering as used in DTP (2 marks)

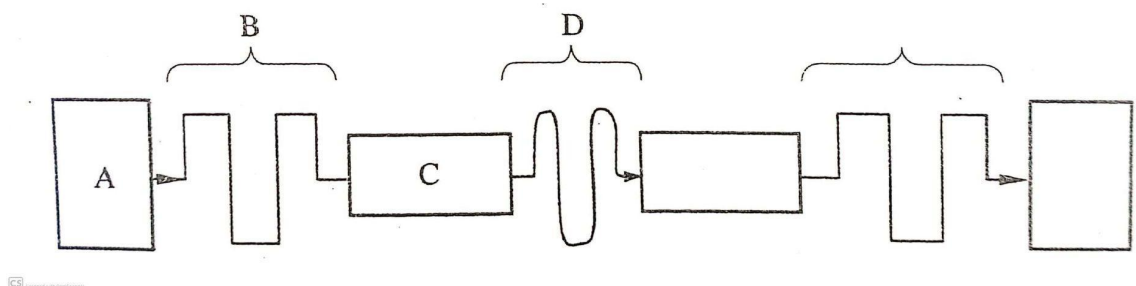
c) Give **two** types of errors that may be detected during program testing. **(2 marks)**

17. a) State **four** functions which are specific to Network Operating Systems.

(4 marks)

b) State **two** advantage of using fibre optic cables over satellite in data communication. **(2 marks)**

c) (i) The figure below shows how data is transmitted through a public telephone . **(4 marks)**



Name A, B, C, and D

d) (i) List any **two** items that may be referred to as data terminal equipments in a network **(2 mks)**

(ii) Write the acronym MODEM in full **(1 mark)**

(iii) Explain the purpose of a modem when connecting to the internet. **(2 marks)**

18. (a) State **three** roles of a programmer in system development life cycle. **(3 marks)**

(b) State **four** activities that may be carried out when disposing off an old system in an

organization

(4 marks)

(c) Distinguish parallel changes over from straight change over as used in system implementation.

(2 marks)

(d) Discuss **any two** fact finding methods.

(4 marks)

(e) Differentiate between an open system and a closed system.

(2 marks)

19. a) State **three** standard coding scheme used computing and electronic systems. **(3 marks)**

b) Convert each of the following numbers

i) 1001.001_2 to octal.

(2 marks)

ii) 125.5_{10} to hexadecimal.

(3 marks)

iii) Add 1100.011_2 to 11001.0101_2 and leave your answer in decimal

(3 marks)

c) Using two compliment perform the following arithmetic leaving your answer in binary form.

(4 marks)

$$13_{10} - 15_{10}$$

20. (a) Distinguish between axis labels and data labels as used in spreadsheets.

(2 marks)

(b) Name an electronic spreadsheet feature that allows one to perform the following actions.

(4 marks)

Action	Feature
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Provide pictorial summary of data	
Make one cell out of many	
List items from Highest to lowest or vice versa	
Make many cells out of one	

(c) Explain the difference between the printing of multiple pages and multiple copies as used in word processing. **(2 marks)**

(d) Describe the following categories of software **(4 marks)**

- i) Firmware
- ii) Proprietary software

(e) Explain the meaning of solid state storage media giving **two** examples of it.

(3 marks)